

Scenario: Student Evening Routine

A student decides every evening how to spend time before an exam. The choices are:

- Study
- Sleep early
- Hang out with friends

The goal is to maximize long-term academic performance and energy.

1. Markov Decision Process (MDP) Design

State Space (S):

$$S = \{F, T, P\}$$

Where:

F = Focused or energetic

T = Tired

P = Prepared for exam

Action Space (A):

$$A = \{\text{study, sleep, hangout}\}$$

Transition Probabilities $P(s'|s,a)$:

From state F (Focused):

study:

- P with probability 0.7
- T with probability 0.3

sleep:

- F with probability 0.8
- T with probability 0.2

hangout:

- T with probability 0.6
- F with probability 0.4

From state T (Tired):

study:

- P with probability 0.4

- T with probability 0.6

sleep:

- F with probability 0.9

- T with probability 0.1

hangout:

- T with probability 0.8

- F with probability 0.2

From state P (Prepared):

Prepared is a stable state:

- Any action leads to P with probability 1.0

Reward Function $R(s,a)$:

study = +10

sleep = +6

hangout = +2

Additional penalty:

- If next state becomes T, reward gets -4.

2. Simplified 3-State Version for Manual Value Iteration

States:

$S = \{F, T, P\}$

Actions:

$A = \{\text{study, sleep}\}$

Discount factor:

$\gamma = 0.9$

Initial values:

$V_0(F) = V_0(T) = V_0(P) = 0$

Value Iteration Formula:

$V(k+1)(s) = \max \text{ over actions of } \sum P(s'|s,a)[R + \gamma * V_k(s')]$

Step 1: Compute V_1

State F:

study:

$$0.7(10) + 0.3(6) = 8.8$$

sleep:

$$0.8(6) + 0.2(2) = 5.2$$

$$V1(F) = 8.8$$

State T:

study:

$$0.4(10) + 0.6(6) = 7.6$$

sleep:

$$0.9(6) + 0.1(2) = 5.6$$

$$V1(T) = 7.6$$

State P:

$$V1(P) = 12$$

After Step 1:

$$V1 =$$

$$F = 8.8$$

$$T = 7.6$$

$$P = 12$$

Step 2: Compute V2

State F:

study:

$$\begin{aligned} &0.7(10 + 0.9 \times 12) + 0.3(6 + 0.9 \times 7.6) \\ &= 18.412 \end{aligned}$$

sleep:

$$\begin{aligned} &0.8(6 + 0.9 \times 8.8) + 0.2(2 + 0.9 \times 7.6) \\ &= 12.904 \end{aligned}$$

$$V_2(F) = 18.412$$

State T:

study:

$$0.4(10 + 0.9 \times 12) + 0.6(6 + 0.9 \times 7.6) \\ = 16.024$$

sleep:

$$0.9(6 + 0.9 \times 8.8) + 0.1(2 + 0.9 \times 7.6) \\ = 13.412$$

$$V_2(T) = 16.024$$

State P:

$$V_2(P) = 12 + 0.9(12) = 22.8$$

Final Values After Two Iterations:

$$V_2 =$$

$$F = 18.412$$

$$T = 16.024$$

$$P = 22.8$$

Interpretation:

- The "Prepared" state has the highest long-term value.
- "Study" becomes the optimal action in both F and T states.
- Value Iteration improves future reward estimates using Bellman updates.